any edge to another vertex within group A, and likewise for group B).

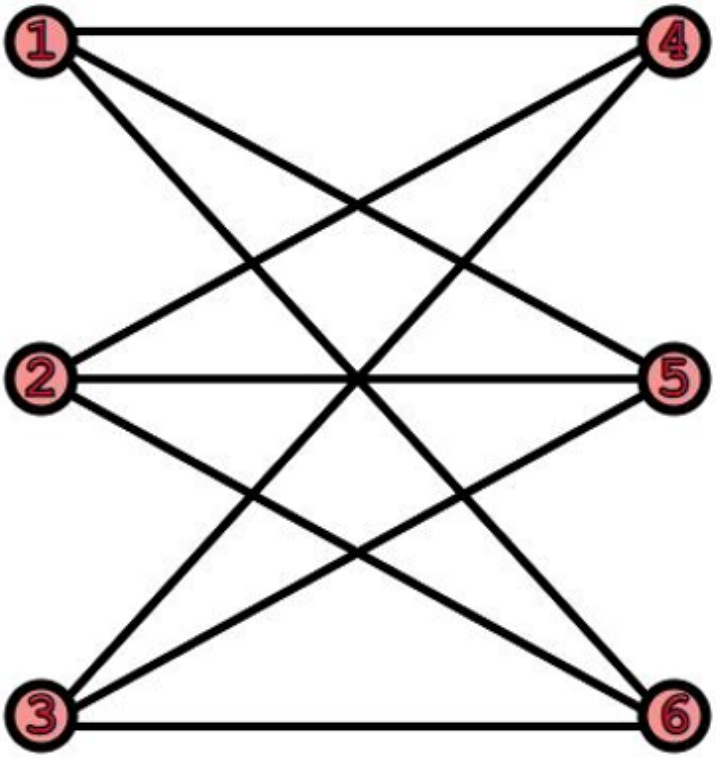


Figure 15.6 An embedding of $K_{3,3}$. Again, do you think there could be a more convenient embedding for this graph?

A **complete bipartite graph** just has all the possible edges between the two groups of vertices – again, loops are not allowed.

So that $K_{n,m}$ is the generic complete bipartite graph with two partitions of n and m vertices each, and $K_{3,3}$ (figure 15.6) is the complete bipartite graph with two partitions having 3 vertices each.